

A Case Study

Diabetes Patient Data Analysis

1. Summary

This case study delves into the Pima Indians Diabetes dataset to uncover the key physiological factors associated with the presence of diabetes. Through a comprehensive process of data cleaning, imputation, and exploratory data analysis (EDA), significant correlations were identified. The analysis reveals that **glucose level is the most potent indicator of diabetes**. Other critical factors, such as Body Mass Index (BMI), age, and the number of pregnancies, also show a strong positive correlation with a diabetes diagnosis. This report details the methodology used to handle missing data—notably using a regression model for Serum Insulin and presents the key findings through a series of targeted questions and visualizations.

2. Data understanding

The dataset consists of 768 patient records with 9 distinct features:

- **Pregnant:** Number of times pregnant
- **Glucose:** Plasma glucose concentration
- **Diastolic BP:** Diastolic blood pressure (mm Hg)
- **Skin Fold:** Triceps skin fold thickness (mm)
- **Serum Insulin:** 2-Hour serum insulin (mu U/ml)
- **BMI:** Body mass index
- **Diabetes Pedigree:** A function scoring the genetic likelihood of diabetes
- **Age:** Age in years
- **Class:** The outcome variable (0 = Non-Diabetic, 1 = Diabetic)

Initial inspection revealed significant missing values in Serum Insulin (48.7% missing) and Skin Fold (29.6% missing).

3. Data Analysis Pipeline

A systematic pipeline was followed to ensure a robust and logical analysis of the dataset. Each step built upon the previous one, leading from raw data to actionable insights.

1. Data Loading and Initial Inspection:

- Imported necessary Python libraries (Pandas, Seaborn, Matplotlib, Scikit-learn).
- Loaded the Diabetes Missing Data.csv dataset into a Pandas DataFrame.
- Performed an initial assessment using `info()` and `skim()` to understand data types, column names, and the quantity of missing values.

2. Data Cleaning and Imputation:

- Addressed missing values in columns with low-to-moderate null counts (Glucose, Diastolic_BP, Skin_Fold) using mean and median imputation strategies.
- Applied a grouped median imputation for BMI based on the Age column to improve accuracy.
- Implemented a regression-based imputation for Serum_Insulin, which had a high percentage (~49%) of missing data. This involved:
 - Separating the dataset into training (known insulin) and prediction (unknown insulin) sets.
 - Training a LinearRegression model.
 - Predicting and filling the missing values with the model's output.

3. Exploratory Data Analysis (EDA) and Visualization:

- Generated descriptive statistics and visualizations to answer
- Utilized histograms to understand the distribution of key variables like Age.
- Created boxplots to compare features (BMI, Age, Pregnancies) across the two Class labels (Diabetic vs. Non-Diabetic).
- Built a correlation heatmap to quantify the relationships between all numerical variables.
- Used scatter plots to visualize the relationship between specific features like BMI and Glucose.

4. Problem Statement

The primary objective of this case study is to analyze patient data to identify the most significant indicators of diabetes. The analysis aims to answer the following key questions:

1. How many missing values are there in each column, and what is the best way to handle them?

Checking for missing values

```
missing_values = D.isnull().sum()
```

```
missing_percentage = (D.isnull().sum() / len(D)) * 100
```

Combine results into a Data Frame

```
missing_df = pd.DataFrame({"Missing Values": missing_values, "Percentage (%)":  
missing_percentage})  
print(missing_df)
```

	Missing Values	Percentage (%)
Pregnant	0	0.000000
Glucose	5	0.651042
Diastolic_BP	35	4.557292
Skin_Fold	227	29.557292
Serum_Insulin	374	48.697917
BMI	11	1.432292
Diabetes_Pedigree	0	0.000000
Age	0	0.000000
Class	0	0.000000

2. What is the age distribution of patients in this dataset?

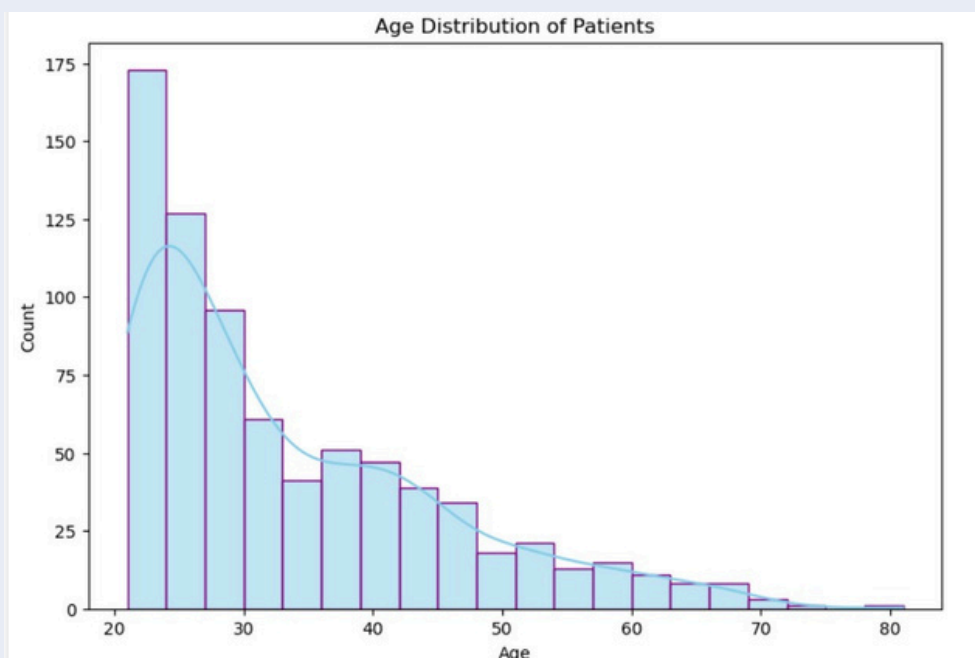
```
plt.figure(figsize=(9,6))
```

```
sns.histplot(diabetes["Age"],bins=20,kde=True,color="skyblue",edgecolor="purple")
```

```
plt.xlabel("Age")
```

```
plt.ylabel("Count ")
```

```
plt.title("Age Distribution of Patients")
```



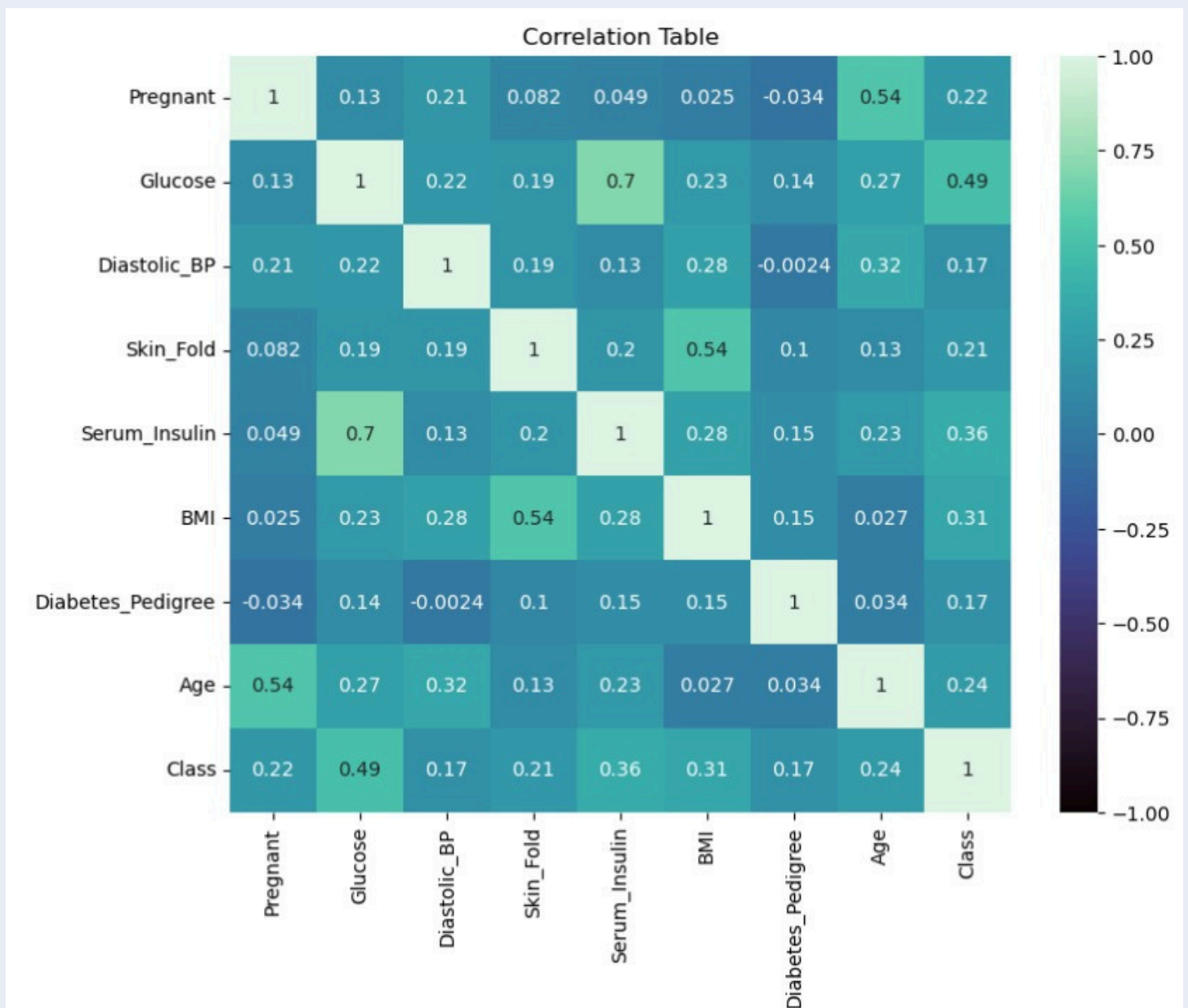
3. What is the correlation between Glucose, BMI, and Diabetes occurrence?

```
plt.figure(figsize=(9,7))
```

```
sns.heatmap(diabetes.corr(),annot=True,cmap="mako",vmax=1,vmin=-1)
```

```
plt.title("Correlation Table")
```

```
plt.show()
```



4. How does BMI vary among diabetic and non-diabetic individuals?

```
plt.figure(figsize=(8,5))

sns.boxplot(x=diabetes["Class"], y=D["BMI"], hue=D["Class"], palette="rocket",
            legend=True)

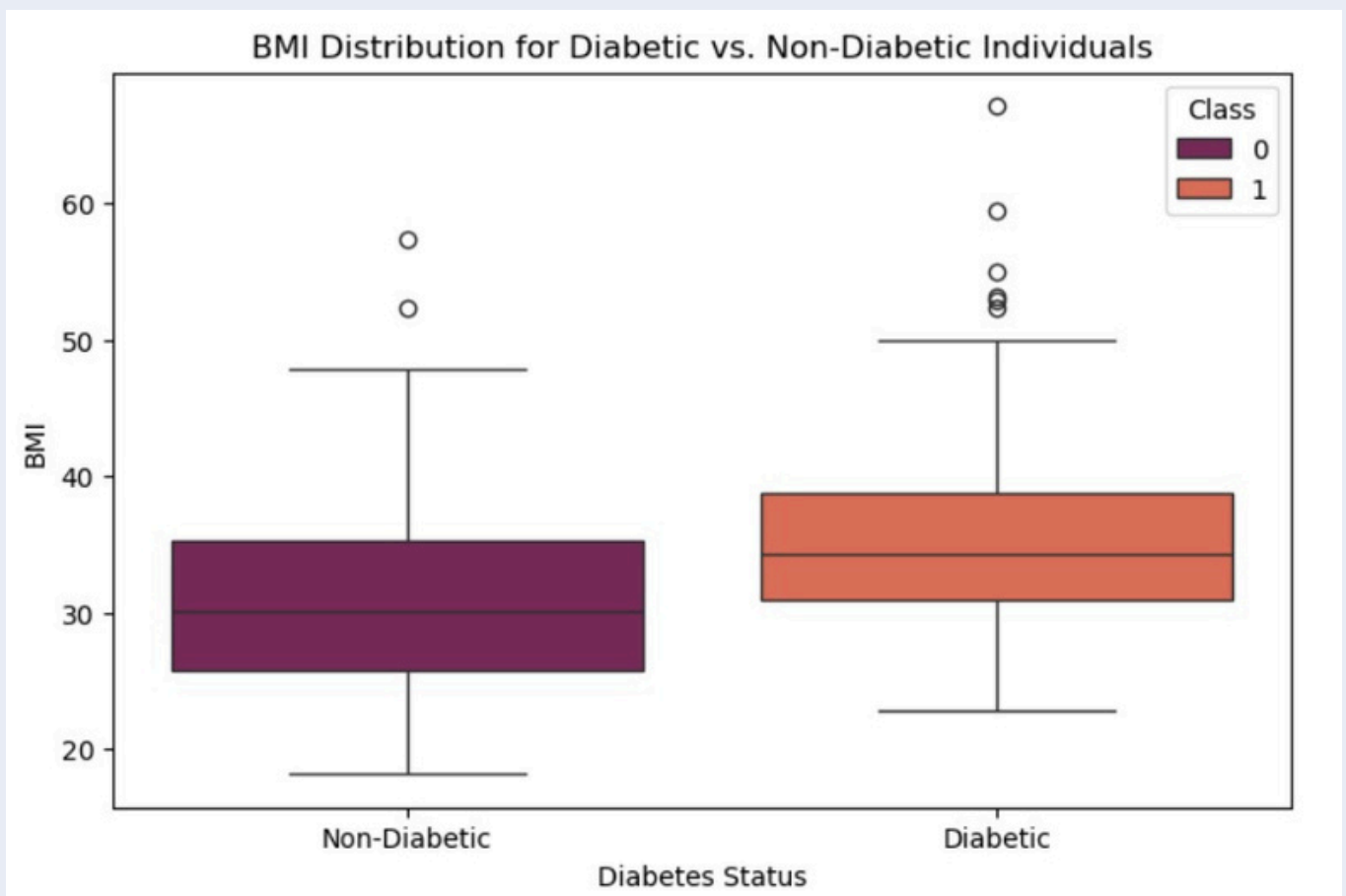
plt.xticks([0,1], ["Non-Diabetic", "Diabetic"])

plt.title("BMI Distribution for Diabetic vs. Non-Diabetic Individuals")

plt.xlabel("Diabetes Status")

plt.ylabel("BMI")

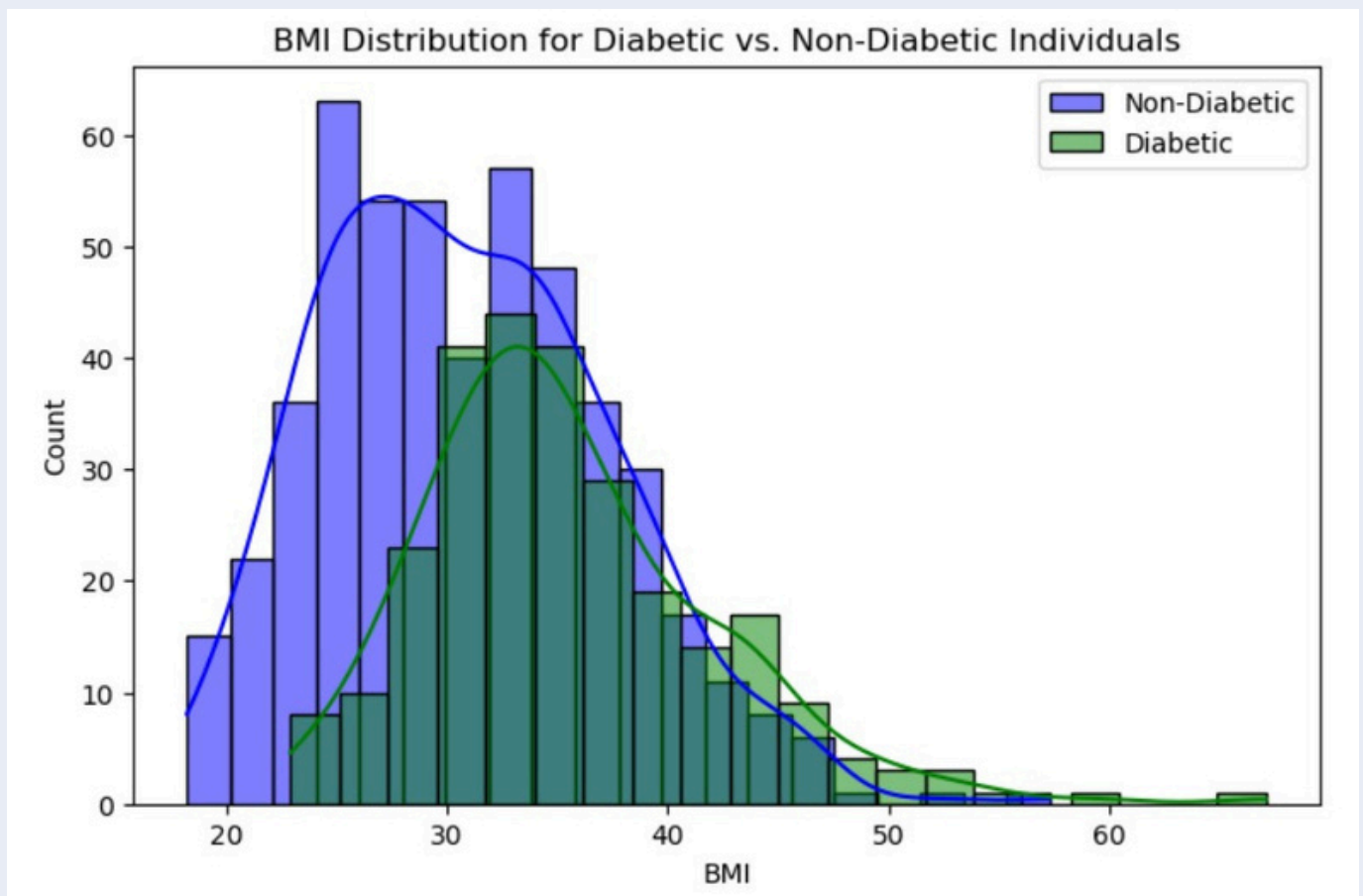
plt.show()
```



5. What are the most common BMI ranges for diabetic and non-diabetic individuals?

```
plt.figure(figsize=(8,5)) sns.histplot(diabetes[diabetes["Class"]==0]
["BMI"],bins=20,color="blue",label="Non-
Diabetic",kde=True)
sns.histplot(diabetes[diabetes["Class"]==1]
["BMI"],bins=20,color="green",label="Diabetic",kde=True)
plt.legend() plt.title("BMI Distribution for Diabetic vs. Non-Diabetic Individuals")

plt.xlabel("BMI") plt.ylabel("Count ") plt.show()
```



5. Exploratory Data Analysis (EDA) & Key Findings

The cleaned dataset was analyzed to answer the stated objectives:

- **Age Distribution:** A histogram of patient ages revealed a right-skewed distribution, indicating that the patient population is predominantly young, with the largest concentration in the 20-30 age bracket.
- **Key Correlations:** The correlation heatmap was highly insightful, showing that **Glucose has the strongest positive correlation with the Class variable**. BMI and Age also showed significant positive correlations with the diabetes outcome.
- **BMI and Diabetes:** Boxplots clearly illustrated that the median BMI for diabetic patients (Class 1) is substantially higher than for non-diabetic patients (Class 0). This suggests a strong link between higher BMI and the likelihood of having diabetes.
- **Age and Diabetes:** Similar to BMI, the age distribution for diabetic patients was higher. The boxplot showed a higher median age and a wider age range for the diabetic group compared to the non-diabetic group.
- **Glucose Levels and Diabetes:** This was the most definitive finding. The average glucose level for diabetic patients was approximately **142**, compared to just **110** for non-diabetic patients. The clear separation in glucose levels between the two groups makes it a primary indicator for diabetes.
- **Pregnancies and Diabetes:** The data showed that diabetic individuals, on average, have a higher number of pregnancies than non-diabetic individuals, indicating a positive correlation.

6. Conclusion

The analysis of the Pima Indians Diabetes dataset successfully identified several key factors strongly associated with a diabetes diagnosis. Higher **BMI**, advanced **Age**, and a greater **number of pregnancies** are all strongly correlated with the presence of diabetes. The careful data cleaning process, particularly the use of regression imputation for Serum Insulin, was critical in ensuring the reliability of these findings.

7. Recommendations

- **Feature Engineering:** New features could be created to potentially improve model accuracy. For instance, creating categorical bins for BMI (e.g., 'Normal', 'Overweight', 'Obese') or Age ('Young', 'Adult', 'Senior') might capture non-linear relationships.
- **Further Analysis:** Statistical tests, such as T-tests, could be performed to confirm if the differences in means (e.g., for Glucose and BMI) between the two classes are statistically significant .